

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$y = 3,600(a)^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y , in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

Correct Answer: 3600

Rationale

The correct answer is **3,600**. The predicted number of bacteria initially measured in the growth medium is the value of y when $x = 0$. Substituting **0** for x in the given equation yields $y = 3,600(a)^0$. Since a is positive, this equation is equivalent to $y = 3,600(1)$. Therefore, when $x = 0$, the value of y is **3,600**. Thus, the predicted number of bacteria initially measured in the growth medium is **3,600**.